N-ATLaS Integration Evidence

N-ATLaS Health QA. A health-sector LoRA adapter for N-ATLaS.

1. What this shows

Every answer from this project comes from N-ATLaS. The artefact is a LoRA adapter: a small set of extra weights that only works when loaded on top of the published N-ATLaS model. No other language model is called anywhere in the code.

- Base model: **NCAIR1/N-ATLaS**, downloaded from Hugging Face under its gated terms.
- Adapter: **Emar7/N-ATLaS-Health-QA-LoRA-Powered-by-Awarri** (https://huggingface.co/Emar7/N-ATLaS-Health-QA-LoRA-Powered-by-Awarri)
- Code: https://github.com/Emart29/emart-health

2. How the adapter loads on top of N-ATLaS

The adapter is loaded with the PEFT library. The call to *PeftModel.from_pretrained* takes the N-ATLaS model object as its first argument. The adapter's own config file records *base_model_name_or_path = NCAIR1/N-ATLaS*.

```
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig
from peft import PeftModel

BASE = "NCAIR1/N-ATLaS"                      # gated: accept the terms on Hugging Face first
ADAPTER = "Emar7/N-ATLaS-Health-QA-LoRA-Powered-by-Awarri"

tok = AutoTokenizer.from_pretrained(BASE)
bnb = BitsAndBytesConfig(load_in_4bit=True, bnb_4bit_quant_type="nf4",
                         bnb_4bit_use_double_quant=True, bnb_4bit_compute_dtype=torch.float16)
base = AutoModelForCausalLM.from_pretrained(BASE, quantization_config=bnb, device_map={"": 0},
                                            torch_dtype=torch.float16)
model = PeftModel.from_pretrained(base, ADAPTER)   # the adapter sits on top of N-ATLaS
model.eval()

system = ("You are a careful medical assistant answering multiple-choice medical exam questions. "
          "Reply with the letter of the single best option.")
user = ("Question:\nWhich investigation should be done first for a febrile child with pallor and a "
        "palpable spleen in a malaria-endemic area?\n\nOptions:\nA. Chest X-ray\n"
        "B. Blood film for malaria parasites\nC. Lumbar puncture\nD. Urine culture\n\n"
        "Answer with the letter of the single best option.")
messages = [{"role": "system", "content": system}, {"role": "user", "content": user}]
prompt = tok.apply_chat_template(messages, add_generation_prompt=True, tokenize=False,
                                 date_string="11 Oct 2026")
enc = tok(prompt, return_tensors="pt", add_special_tokens=False).to(model.device)
out = model.generate(**enc, max_new_tokens=8, do_sample=False)
print(tok.decode(out[0, enc["input_ids"].shape[1]:], skip_special_tokens=True))
```

3. Prompt template

Prompts are built with N-ATLaS's own chat template (*tokenizer.apply_chat_template*), so the special tokens are exactly the ones the base model was trained with. The same template is used for training, for scoring base N-ATLaS and for scoring the fine-tuned model.

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

[N-ATLaS default header with the date string]

You are a careful medical assistant answering multiple-choice medical exam questions. Reply with the
letter of the single best option.<|eot_id|><|start_header_id|>user<|end_header_id|>

Question:
{question}

Options:
```

```
A. {option 1}
B. {option 2}
...

Answer with the letter of the single best option.<|eot_id|><|start_header_id|>assistant<|end_header_id|>

{answer letter}
```

4. Training setup

Setting	Value
Base model	NCAIR1/N-ATLaS (Llama architecture, 8 billion parameters), loaded in 4-bit NF4
Method	QLoRA: the base weights stay frozen, only low-rank adapter weights are trained
LoRA	rank 16, alpha 32, dropout 0.05, on all attention and MLP projection layers
Trainable parameters	41,943,040 trainable parameters, which is 0.52% of the 8.07 billion parameters in the model
Training examples	3222 single-answer multiple-choice questions
Epochs / learning rate	1 / 0.0002 with a short linear warmup, then cosine decay
Batch	2 per GPU on 2 GPUs with 8 gradient accumulation steps, so an effective batch of 32. 101 optimizer steps, the first 10 as warmup
Loss	Only on the answer (the option letter, plus the expert rationale where the dataset has one)
Hardware and time	Kaggle free tier, 2 x NVIDIA Tesla T4 (16 GB each). Training took 44 minutes. The whole run, including both benchmark passes, took about 51 minutes
Seed	42

5. Sample output

Question put to N-ATLaS with the adapter loaded, at the end of the training run:

```
A 4-year-old child in a malaria-endemic area presents with fever, pallor and a palpable spleen.
Which investigation should be done first?

A. Chest X-ray
B. Blood film for malaria parasites
C. Lumbar puncture
D. Urine culture
```

Model output, copied from the run log:

```
B
```

6. Result on the 500-question benchmark

Model	Correct	Accuracy	95% confidence interval (Wilson)
Base N-ATLaS (NCAIR1/N-ATLaS)	252 / 500	50.4%	46.0% to 54.8%
N-ATLaS + this adapter	303 / 500	60.6%	56.2% to 64.8%
Chance (random guessing)		21.2%	

The fine-tuned model was more accurate than base N-ATLaS by 10.2 percentage points on the same 500 questions. The exact McNemar p-value is 1.4 x 10^-6, so the difference is statistically significant at the 0.05 level.

7. Licence and attribution

The adapter is a derivative of N-ATLaS and is released under the same N-ATLaS Terms of Use (research and education use, capped at 1,000 active users). Its name carries the required suffix "Powered by Awarri".

N-ATLaS is an initiative of the Federal Ministry of Communications, Innovation and Digital Economy, and powered by Awarri Technologies.

Training and test data: AfriMed-QA v2 (CC BY-SA 4.0), Olatunji et al., 2024, arXiv:2411.15640.

This is a research model. It is not a medical device and must not be used for diagnosis or treatment.